

Intro to Software Engineering

Final Project Report

NUDTEE

A web application for finding board game sessions and coordinating participants

Team Dek70

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Repository: <https://github.com/6931503074-sudo/NUDTEE>

Prototype / demo: <https://nudtee-3f5b7.web.app/>

1. Problem and Target Users

NUDTEE is a web application that helps board game players find suitable game sessions and coordinate the people who join them. Organising a session through group chats creates four documented problems:

1. Players do not know which game genres the other members of a group enjoy.
2. Organisers have difficulty finding participants who are interested in their session.
3. The number of participants in a session is difficult to maintain through chat messages.
4. Players may miss the opportunity to join a session that suits them.

The target users are board game players who look for sessions to join, and organisers who create sessions and need to know how many people have joined.

Success measure and project scope (M1):

ID	Metric	Target
SM-1	Time for an organizer to create a session	60 seconds or less
SM-2	Taps for a player to join an open session from the list	2 taps or fewer
SM-3	Time for another device to show a join/leave change 2 seconds or less	2 seconds or less
SM-4	Filtered results that match the selected genre	100%

2. Requirements Summary

Functional requirements and priorities

ID	Requirement	Priority
FR-1	Create a session.	Must
FR-2	Set favorite game genres.	Must
FR-3	Join or leave a session.	Must
FR-4	Filter sessions by genre.	Should
FR-5	Display real-time session status.	Must

Non-functional requirements and acceptance criteria:

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	The session list shall reflect a join/leave change on all connected clients within 2 seconds.	Two devices open the same session list; one joins, time until the other updates
NFR-2	usability	The app shall be usable on a mobile phone screen without horizontal scrolling on any core screen.	Manual check on a real phone for each of the 5 core screens
NFR-3	capacity	The system shall support at least 20 concurrently connected users viewing the real-time session list without errors.	Load test with 20 simulated concurrent connections
NFR-4	reliability	A session's remaining-spot count shall never go negative or exceed the stated maximum, even under concurrent join attempts.	Two clients tap join simultaneously on a session with one spot

ID	Kind	Statement (testable)	How we will check
			left; verify only one succeeds
NFR-5	security / privacy (required)	A user shall be able to edit or delete only sessions they created; attempting to modify another user's session shall fail.	Attempt to edit another account's session via the API directly and confirm it is rejected

3. System Design and Diagrams

3.1 Architecture

Three-tier architecture

Tier	Components	Requirement
Presentation	Session list screen: open sessions with genre, location, time and a live remaining-spot count; genre filtering. Create session screen: form for a new session. Profile screen: player selects favorite genres.	FR-4, FR-5; FR-1; FR-2

Tier	Components	Requirement
Application logic	Create session: validates and stores a new session. Set favorite genres: validates and stores genre tags. Join session: checks capacity, rejects if full, adds the participant. Leave session: removes the participant and recalculates remaining spots. Filter sessions: returns sessions matching the selected genres. Broadcast session update: notifies connected viewers when the participant count changes.	FR-1; FR-2; FR-3; FR-3; FR-4; FR-5
Data	Player, Session and Participation entities (Table 3).	FR-1 to FR-5

3.2 Data Model

Entities, fields and relationships

Entity	Fields	Relationships	Used by
Player	id, display name, favorite genres (list)	Organises many Sessions; has many Participations	FR-2
Session	id, organizer (Player), genre, location, scheduled time, minimum players, maximum players, status	Belongs to one organizing Player; has many Participations	FR-1, FR-4, FR-5

Entity	Fields	Relationships	Used by
Participation	session (Session), player (Player), joined-at time	Belongs to one Session and one Player	FR-3

3.3 Diagrams

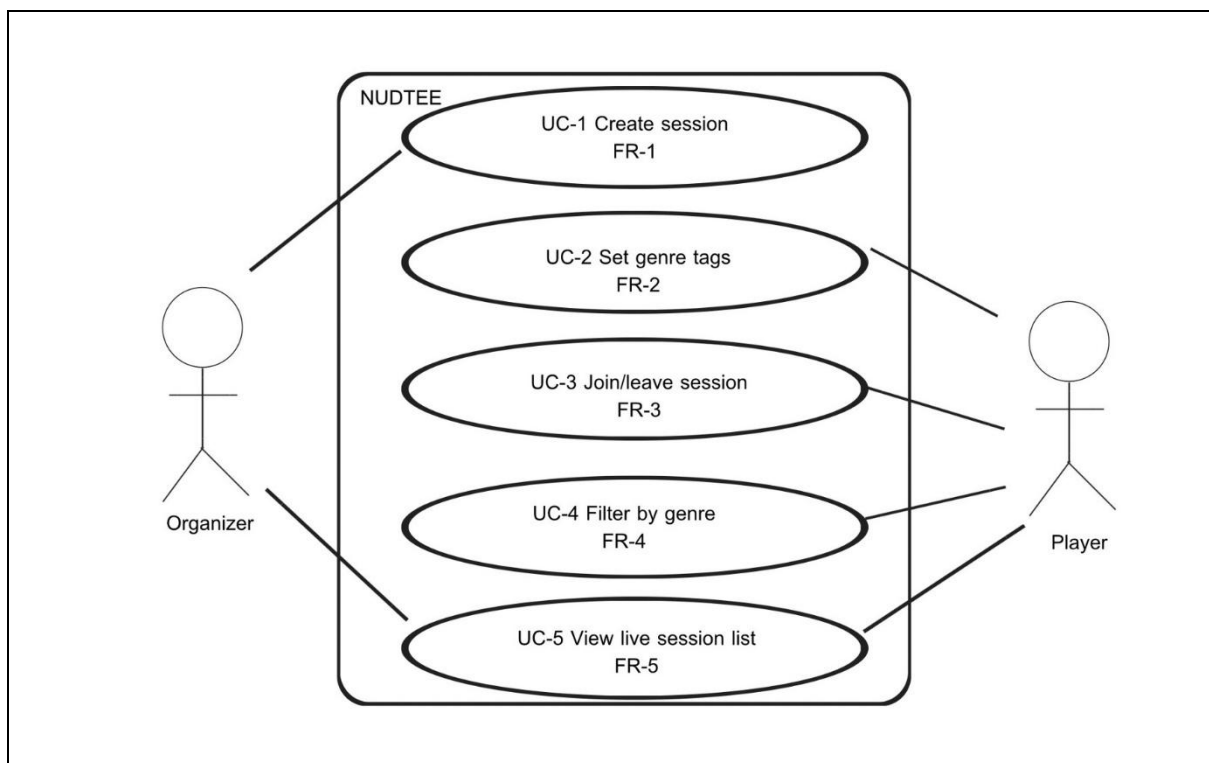


Figure 1. Use Case diagram of NUDTEE

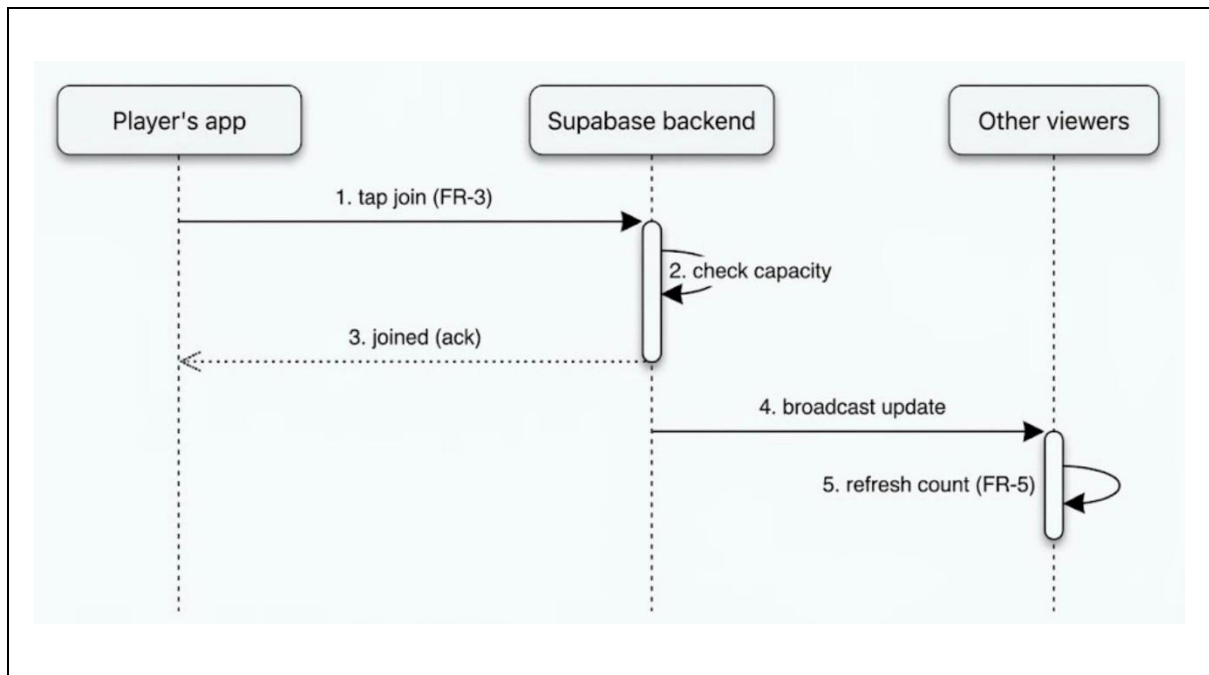
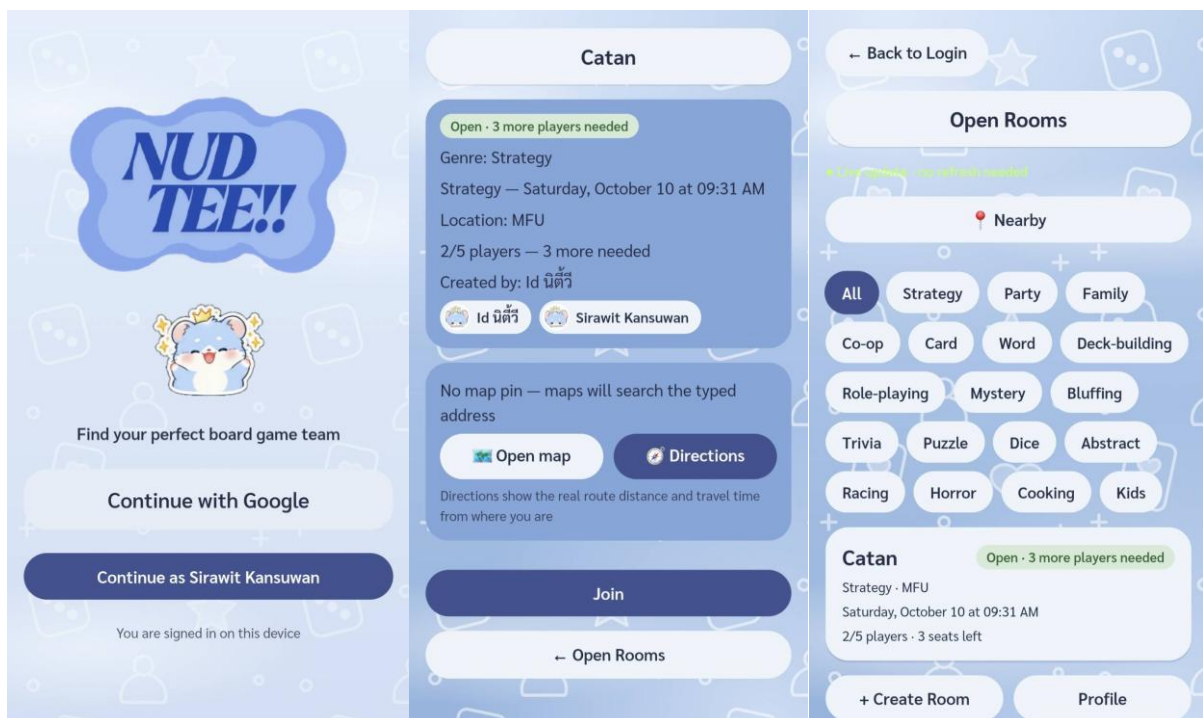


Figure 2. Sequence diagram of NUDTEE

4. Prototype and Implementation

Tools used and prototype link: Hosting:Firebase Prototype: <https://nudtee-3f5b7.web.app/>



+ Create Room

Game name

Enter game name...

Game Genre

Select a genre

Location

Enter location...

Pin meeting spot (use my location)

Optional — lets nearby players find your room

Date & Time

ว/ดศ/ปปปป --:--

Minimum Players

e.g. 3

Maximum Players

e.g. 5

Create room

My Profile

Sirawit Kansuwan

Name

Sirawit Kansuwan

About me

e.g. Loves co-op games

Preferred Game Genres (Multiple selections)

Strategy

Party

Family

Co-op

Card

Word

Game name

Catan

Game Genre

Co-op

Location

MFU

Pin meeting spot (use my location)

Optional — lets nearby players find your room

Date & Time

10/10/2026 09:12

Minimum Players

5

Maximum Players

3

Maximum players cannot be less than minimum

Create room

Open Rooms

Catan

Full

Genre: Strategy

Strategy — Saturday, October 10 at 09:13 AM

Location: MFU

2/2 players

Created by: Id นิตวิ

No map pin — maps will search the typed address

Open map

Directions

Directions show the real route distance and travel time from where you are

Join

Open Rooms

Live update - no refresh needed

Nearby

All

Strategy

Party

Family

Co-op

Card

Word

Deck-building

Role-playing

Mystery

Bluffing

Trivia

Puzzle

Dice

Abstract

Racing

Horror

Cooking

Kids

No rooms yet — tap + Create Room to make the first one

+ Create Room

Profile

Status of each requirement

ID	Planned (SRS)	Designed	Prototype	Implemented	Tested
FR-1	Yes	Yes	Yes	Yes	Yes
FR-2	Yes	Yes	Yes	Yes	Yes
FR-3	Yes	Yes	Yes	Yes	Yes
FR-4	Yes	Yes	Yes	Yes	Yes
FR-5	Yes	Yes	Yes	Yes	Yes

5. Testing and Results

Planned test scenarios and results

Test	FR	Scenario (planned)	Expected result (from design)	Actual result
TC-1	FR-1	An organizer submits a valid new session.	The session is validated, stored and appears in the session list.	Pass
TC-2	FR-2	A player selects favorite genres on the profile screen.	The genre tags are validated and stored for the player.	Pass
TC-3	FR-3	A player joins a session that has remaining spots.	The participant is added and the remaining-spot count decreases.	Pass
TC-4	FR-3	A player joins a session that is full.	The join is rejected.	Pass

Test	FR	Scenario (planned)	Expected result (from design)	Actual result
TC-5	FR-3	A participant leaves a session.	The participant is removed and the remaining spots are recalculated.	Pass
TC-6	FR-4	A player selects a genre filter.	Only sessions matching the genre are shown.	Pass
TC-7	FR-5	The participant count of a session changes.	Every connected viewer is notified of the update.	Pass

6. AI-Use Reflection

This section will summarise the team's AI-Use Statement. It will distinguish tasks where AI assisted the team, decisions made and verified by team members, AI-generated material that the team reviewed or corrected, and the limitations and mistakes encountered.

Tool used: Claude (Claude Sonnet 5), via chat. Used throughout M1–M3 for drafting and review; not used to write code or commit to the repository directly. Where AI helped: - Scoping the problem and MVP (M1) — turning a vague idea into a specific problem statement, identifying who experiences the problem, and drafting a first-pass MVP feature list. - Drafting the SRS (M2) — writing FR-1 to FR-5 as user stories with acceptance criteria, flagging which FRs have real decision logic (FR-3's capacity check, FR-5's realtime sync) versus simple CRUD, and drafting the NFR table, use-case list, and MoSCoW/INVEST tables for the team to build on. - Diagrams and design (M3) — first drafts of the use case diagram, ERD, sequence diagram, a tech-agnostic architecture document (three tiers + data model), and a design-token set (design-system.md). - Prototype scaffolding (M3) — a clickable HTML/CSS/JS prototype

covering every Must FR, built from the design tokens above, including unhappy-path states (full session, invalid form input, empty filter result). What worked well: - Fast, reasonable first drafts meant the team spent its time deciding and refining rather than starting from a blank page. - AI was useful for flagging gaps against the course rubric — e.g. noticing the SRS was missing MoSCoW priorities and edge-case acceptance criteria before the team submitted the revised version. - Reusing one consistent set of design tokens across every diagram, wireframe, and the final prototype was easier because AI reused the same values every time. What didn't work / needed correction: - AI's first-pass acceptance criteria only covered the happy path; FR-1, FR-2, and FR-4 needed an explicit second request before edge/error cases were added. - An unverified claim about Supabase/Google/Vercel free-tier limits was initially presented with confidence; the team had Claude re-check it with a live web search rather than trusting the first answer. - AI cannot act outside the chat — it could not verify that the GitHub repo was actually public or push commits; the team checked all of this directly in a browser and terminal. - AI-drafted text needed hand-editing for accuracy, not just polish — an AI-usage-log entry in an earlier SRS draft had been copied from the instructor's own worked example and not edited to describe the team's actual project; the team caught this and rewrote the log to reflect what was actually asked, received, and changed. What the team decided or verified itself: the final problem wording, the MVP boundary (Must/Should/Could/Won't), the app's name (NUDTEE), every entity field name and design-token value, team role assignments, and this statement itself were reviewed and decided by the team — not generated and accepted as-is.

7. Individual Contributions

Member responsibilities

Student ID	Name	Responsibility	links
6931503050	Ninmanee Nualsee	Prototype,design,Test result	https://github.com/6931503074-sudo/NUDTEE/commits/main/?author=Ninmanee
6931503065	Rawisara Palang	Prototype,design, Test result	https://github.com/6931503074-sudo/NUDTEE/commits/main/?author=6931503065-web
6931503119	Suratsawadee Areerob	SRS,diagram,final report	https://github.com/6931503074-sudo/NUDTEE/commits/main/?author=sSuratreerob
6931503074	Sirawit Kansuwan	Scope,planning,SRS,final report	https://github.com/6931503074-sudo/NUDTEE/commits/main/?author=6931503074-sudo
6931503107	Thanyarat Ainpasat	Diagrams,SRS,AI usage	https://github.com/6931503074-sudo/NUDTEE/commits/main/?author=6931503107-sketch

Due to time constraints, most commits were pushed directly to the main branch via the GitHub web interface. Therefore, individual commit links are used as evidence of individual contributions instead of pull request (PR) links.

8. Golden Thread Traceability Table

Problem → Requirement → Design → Feature → Test

Problem	Requirement	Design	Feature (screen)	Test
Players do not know which genres others enjoy.	FR-2	Set favorite genres operation; Player entity	Profile screen. Status:Pass	TC-2. Result: Pass
Organisers have difficulty finding interested participants.	FR-1	Create session operation; Session entity	Create session screen. Status: Pass	TC-1. Result: Pass
Organisers have difficulty finding interested participants.	FR-4	Filter sessions operation; Session entity	Session list screen (genre chips). Status:Pass	TC-6. Result: Pass

Problem	Requirement	Design	Feature (screen)	Test
Participant counts are difficult to maintain in chat.	FR-3	Join and leave session operations; Participation entity	Session list screen (Join button). Leave screen:Pass. Status: Pass	TC-3, TC-4, TC-5. Result: Pass
Participant counts are difficult to maintain in chat.	FR-5	Broadcast session update operation; Session and Participation entities	Session list screen (live count). Status: Pass	TC-7. Result: Pass
Players may miss suitable sessions.	FR-4, FR-5	Filter sessions; Broadcast session update	Session list screen. Status: Pass	TC-6, TC-7. Result: Pass

9. Relevant Links and References

Repository: <https://github.com/6931503074-sudo/NUDTEE>

Architecture: <https://github.com/6931503074-sudo/NUDTEE/blob/main/Docs/architecture.md>

Traceability: <https://github.com/6931503074-sudo/NUDTEE/blob/main/Docs/traceability.md>

Use Case diagram: <https://github.com/6931503074-sudo/NUDTEE/blob/main/Docs/diagrams/use-case.jpg>

Sequence diagram: <https://github.com/6931503074-sudo/NUDTEE/blob/main/Docs/diagrams/sequence.jpg>

README: <https://github.com/6931503074-sudo/NUDTEE/blob/main/README.md>

Prototype, demo: <https://nudtee-3f5b7.web.app/>